PES UNIVERSITY, Bangalore

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Department of Computer Science & Engineering

UE24CS251A: DIGITAL DESIGN AND COMPUTER ORGANIZATION

Unit 4

1. Derive the expression for the total number of AND gates and Full Adders in an $n \times n$ array multiplier.

2. Explain how the array multiplier implements binary multiplication using digital logic.

3. Perform binary multiplication using the Shift–Add algorithm for M = 1011 (11) and Q = 1101 (13).

4. What is Booth's Algorithm? Explain its operation with bit transitions (Q_0, Q_{-1}) and show how +1, –1, and 0 operations are determined

5. For a multiplier pattern 01111110, demonstrate why it represents the best case for Booth's algorithm. For 010101010, show why it is the worst case (maximum transitions).

6. Explain the Restoring Division Algorithm with steps and an example.

7. Explain the Non-Restoring Division Algorithm and how it improves efficiency.

8. Differentiate between Restoring and Non-Restoring Division Algorithms.

9. For n-bit operands, analyze the time complexity of restoring vs. non-restoring division.

10. Explain how sign handling is performed in binary division of signed numbers.

11.Explain the concept of Floating Point Representation.

12. What are the differences between Fixed Point and Floating Point representation?

13. What are Floating Point Special Cases?

14. What are Floating Point Rounding Modes?

15. Explain the working of a register bit implemented using a D flip-flop, multiplexer, and tri-state gate.

16. What are the control signals used for register transfer operations? Explain their purpose with an example.

17. explain the functional roles of MAR, MDR, and internal processor bus during instruction execution.

18. explain the control sequence for executing a complete instruction in a single-cycle datapath.

19. explain how a branch instruction is executed, and differentiate between unconditional and conditional branches.

20. What is Multiple Bus Organization? How does it improve data transfer speed within the processor?

21. Explain the working of a Hardwired Control Unit. What are its key components and control signals?

22. Compare and contrast Hardwired Control and Microprogrammed Control.

23. Explain the concept of Microprogrammed Control. What is a Control Word (CW)?

24. Distinguish between Horizontal and Vertical Microprogramming.